Engineering Mini-Projects

Engineering Computation with Python — Chapter 16

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1. Circuits: RC, RLC, and a Three-Loop Solver
2. Data: Calibration, Noise, and Prediction
3. Capstones: Energy and the Complete Bench Analysis

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About These Projects

Every chapter of this course taught one technique at a time. Real engineering problems do not arrive sorted by technique: a single question usually needs loading data, plotting it, cleaning it, fitting a model, and interpreting the result — all at once.

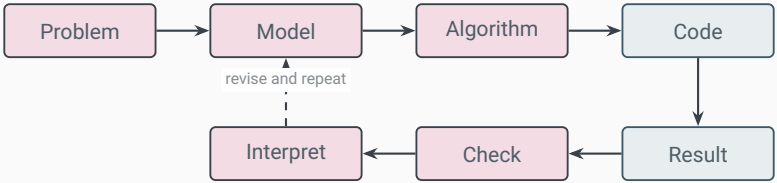
The eight projects ahead are deliberately small, but each combines several chapters. None has a single correct program; what matters is that the answer is right, that you can justify it, and that you have checked it.

Work through the full cycle for every project

Do not skip the checks — a result you have not verified is not yet an answer.

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The Engineering Computation Cycle, Once More



Each project below ends with a short list of checks — a units check, a magnitude check, and an interpretation question. Use them the same way you have all course.

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Chapters used: Plotting, Arrays, Units.

A capacitor C charges through a resistor R from a supply V_0:

$$v_C(t) = V_0\left(1 - e^{-t/\tau}\right), \qquad \tau = RC.$$

Plot $v_C(t)$ for $R = 4.7\,\text{k}\Omega$, $C = 100\,\mu\text{F}$, $V_0 = 12\,\text{V}$, over 0 to 5τ. Repeat on the same axes for $R = 1\,\text{k}\Omega$ and $R = 10\,\text{k}\Omega$. Describe what changing R does to the shape.

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Units

R in ohms times C in farads gives τ in seconds. Here $\tau = 0.47\,\text{s}$.

Limiting case

At $t = \tau$, voltage should reach about 63% of V_0, roughly 7.6 V. At $t = 5\tau$, within 1% of V_0.

Interpret

Does a larger R make the capacitor charge faster or slower? Explain physically, not just from the graph.

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Chapters used: Complex impedance, Log plots.

Series RLC circuit: $R = 40\,\Omega$, $L = 0.2\,\text{H}$, $C = 50\,\mu\text{F}$.

$$Z(f) = R + j\omega L - \frac{j}{\omega C}, \qquad \omega = 2\pi f.$$

Compute $Z(f)$ from 1 Hz to 1 kHz. Plot $|Z|$ and phase θ against frequency on a logarithmic frequency axis. Identify the resonant frequency from your plots.

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Theory

Resonance occurs when $X_L = X_C$, giving $f_0 = 1/(2\pi\sqrt{LC}) \approx 50.3\,\text{Hz}$. Does your magnitude plot reach its minimum there?

Magnitude

At resonance, $|Z|$ should fall to approximately $R = 40\,\Omega$, and the phase should cross zero.

Interpret

Well below resonance the circuit is capacitive (negative phase); well above it is inductive. Confirm this from your phase plot.

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Chapters used: Matrices, Files.

Extend the two-loop mesh solver to three loops. Build the 3×3 coefficient matrix **A** and source vector **b** for resistors and sources of your own choosing, solve $\mathbf{Ax} = \mathbf{b}$ with `np.linalg.solve`, and print all three loop currents in milliamps.

```
def solve_loops(A, b):
    return np.linalg.solve(A, b)   # works for any number of loops
```

Project 6: Checks

Substitution

Compute `A @ x` and confirm it reproduces `b`.

Magnitude

For a low-voltage supply and resistors of hundreds of ohms, currents should be in the milliamp range. Amps would signal a units or matrix-construction error.

Interpret

If a computed current is negative, what does that mean about the direction you assumed for that loop?

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Chapters used: Curve fitting, Plotting.

A temperature sensor outputs a voltage. Calibration data:

Voltage (V)	0.5	1.0	1.5	2.0	2.5	3.0
Temp. (°C)	−2.1	12.4	27.0	41.8	56.2	70.9

Fit $T = mV + c$ with `np.polyfit`, print the equation, plot the data with the fitted line, and compute the residual at each point. Convert a new reading of 1.75 V to temperature.

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Fit

Should give approximately $T = 29.2V - 16.8$.

Residuals

All small (under $0.2\,^{\circ}\mathrm{C}$), no systematic pattern. A curved residual pattern would mean a straight line is the wrong model.

Interpret

What does slope m mean physically, in degrees per volt? What does intercept c represent?

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Chapters used: Arrays, Statistics, Plotting.

Generate a noisy 100 Hz sine wave over two cycles:

$$x(t) = \sin(2\pi f t) + n(t),$$

with $n(t)$ from `np.random.normal`, standard deviation 0.3. Plot the noisy signal, compute its mean and standard deviation, then threshold: set every sample whose magnitude is below 0.5 to zero. Plot the result alongside the original.

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Magnitude

The mean of the noisy sine should be close to zero — a full number of cycles of a sine averages out.

Interpret

Thresholding removed small values — but did it remove only noise? Examine what happened near the zero-crossings of the sine, where the true signal is also small. What does this tell you about the limits of naive filtering?

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Chapters used: Machine learning, Statistics.

Using a small dataset relating motor operating current and ambient temperature to motor body temperature (plausible data, or your own measurements), train a linear regression model with scikit-learn to predict motor temperature from the other two variables.

Split the data into training and testing sets, fit the model, and report its accuracy on the test set.

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Evaluation

Evaluate on the test split, never on the training data.

Magnitude

Predicted temperatures should be physically plausible. A prediction of $-40\,^{\circ}\mathrm{C}$ or $500\,^{\circ}\mathrm{C}$ signals a problem with the data or the model.

Interpret

Does the model say higher current raises motor temperature? Does that match the physics you expect? Distrust a model whose coefficients contradict physical reasoning, even if its accuracy looks good.

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Chapters used: Differentiation/integration, Files.

A battery under a constant 0.5 A load is logged once per minute for one hour; terminal voltage falls approximately linearly from 12.6 V to 11.2 V.

1. Plot voltage against time.
2. Compute dv/dt with `np.gradient`.
3. Compute delivered energy $E = \int v\, i\, dt$ with `np.trapezoid`, converted to watt-hours.

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Units

Volts $\times$ amps $\times$ seconds gives joules; divide by 3600 for watt-hours.

Magnitude

Energy should be roughly 5.9 Wh. Hand check: average voltage 11.9 V $\times$ 0.5 A $\times$ one hour $\approx$ 6 Wh — agrees.

Interpret

dv/dt is negative and roughly constant. What would a sudden steepening of that slope indicate about the battery?

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Chapters used: nearly all of them.

Return to `bench_rc.csv`, the recurring dataset from the start of the course, and carry out the whole workflow yourself:

1. Load the file; inspect shape and columns.
2. Plot voltage against time.
3. Repair the three missing samples near $t = 0.8$ s by interpolation.
4. Detect the corrupted temperature reading with the IQR rule, and replace it.
5. Fit the discharge for the measured time constant.
6. Differentiate the repaired voltage to estimate current from $i = -C\,dv/dt$; compare against the recorded column.
7. Report measured τ, nominal $\tau = RC = 0.47$ s, and the percentage error.

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Project 8: A Hint and the Checks

Hint

Taking the natural logarithm of $v(t) = V_0 e^{-t/\tau}$ gives a straight line whose slope is $-1/\tau$, so `np.polyfit` on `np.log(v)` versus `t` works directly.

Magnitude

Fitted τ should come out near 0.47 s — within a few percent is a good result for noisy measurements. The differentiated current will be noisier than the recorded column; differentiation amplifies noise.

Interpret

Which current figure would you trust more for a report — the measured column, or the one derived from the voltage? Justify your choice.

If you can complete Project 8 unaided, you have done what this course set out to teach.

What that means

You have taken a real engineering measurement, expressed the physics mathematically, translated that mathematics into Python, computed a result, checked it, and interpreted what it means.

That is engineering computation.

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